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Original research

Tool in lesion verification of shape-sensing robotic-assisted bronchoscopy with cone beam CT in sampling peripheral pulmonary nodules

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► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/thorax-2025-223631>).

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Received 31 May 2025
Accepted 6 November 2025
Published Online First
14 December 2025

ABSTRACT

Introduction Our aim was to evaluate the diagnostic and safety performance of shape-sensing robotic-assisted bronchoscopy (ssRAB) with cone beam CT (CBCT) for the biopsy of pulmonary nodules. Additional analysis was performed to assess outcomes for small nodules and those close to the fissure, pleura or mediastinum.

Methods This single arm, multicentre prospective study enrolled 200 subjects with suspicious pulmonary nodules. Each subject underwent a ssRAB procedure with CBCT and was subsequently followed up. The primary outcome was tool-in-lesion (TIL) confirmed with CBCT. Further endpoints included diagnostic outcomes and rate of adverse events.

Results Of 200 subjects recruited, 198 subjects had a successful biopsy (whereby lesion was reached and sample was taken) and 97% completed the required follow-up. The median size of the nodules was 13 mm; 26.8% (60/224) have a positive bronchus sign and 181 (80.8%) were located in the outer two-thirds of the lung. TIL was obtained in 99.0% (198/200). The strict diagnostic yield was 85% (170/200) with diagnostic accuracy of 92.0% (184/200) and sensitivity for malignancy of 95.5% (147/154). Diagnostic accuracy for nodules under 20 mm size, within 5 mm from a critical structure (eg, heart, aorta or main pulmonary artery) or from the pleura was 88.2% (172/195), 100% (11/11) and 93.3% (56/60), respectively. There were four (2%) serious procedure-related adverse events, with one patient (0.5%) suffering a pneumothorax.

Conclusion ssRAB with CBCT can effectively reach and biopsy small pulmonary nodules, including perifissural, peripleural and paramediastinal lesions with a strong safety profile.

Trial registration number NCT05867953.

INTRODUCTION

With the advent of lung cancer screening and increasingly widespread use of CT imaging, there has been a rise in the number of pulmonary nodules identified and investigated. It has been reported that pulmonary nodules were detected in 50% of baseline screening CT scans and up to 27% of non-screening CT scans.^{1–3} Transthoracic needle biopsy (TTNB) has been the mainstay for sampling pulmonary nodules; however, it has been limited by the availability of interventional radiologists and by

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Shape-sensing robotic-assisted bronchoscopy (ssRAB) is a novel biopsy technique whose role in Europe remains to be established. There is minimal data on its diagnostic performance for small nodules and those located near fissures, pleural surfaces or the mediastinum.

WHAT THIS STUDY ADDS

⇒ The study confirms that ssRAB with cone beam CT (CBCT) can successfully reach and biopsy small (median 13 mm) pulmonary nodules, demonstrating a strict diagnostic yield of 85% and diagnostic accuracy of 92.0%. This trial confirms that it has reasonable diagnostic outcomes for smaller nodules and those located close to the pleura or a critical structure, with an excellent safety profile.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ To our knowledge, this is the largest European and only multicentre prospective trial to date evaluating the diagnostic and safety outcomes of ssRAB with CBCT.

nodule characteristics which may increase the rate of complications.^{4–6} The risk of pneumothorax is greater with TTNB and increases by 11-fold in nodules 20 mm or smaller, and by sevenfold if located between 1 mm and 20 mm from the pleura.⁶

Data from the Dutch-Belgian Randomized Lung Cancer Screening Trial (NELSON study) found that over 60% of screening detected cancers were located in the outer third of the lung and 46.8% were diagnosed at stage I (ie, 30 mm or less).^{7,8} There were no reported figures on proximity to major vascular structures; however, it would be entirely plausible to infer that there will be an increasing number of these 'difficult-to-access' nodules detected in screening.⁹ It has been suggested that transthoracic biopsy should be avoided if the nodule is within 10 mm of the pericardium due to cardiac motion.¹⁰

The combination of the above issues with an anticipated rise in pulmonary nodule case volume necessitates a greater repertoire of reliable biopsy techniques to achieve timely and accurate diagnoses.



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To cite: Chan LT, Lau KKW, Orton CM, et al. *Thorax* 2026;**81**:571–580.



As a result, there is increasing recognition for the role of navigation bronchoscopy in biopsying pulmonary nodules. Electromagnetic navigation bronchoscopy (EMN) has been limited by its diagnostic yield with the Clinical Evaluation of superDimension™ Navigation System for Electromagnetic Navigation Bronchoscopy (NAVIGATE) multicentre cohort study reporting a diagnostic accuracy of 73%.¹¹ There has been growing evidence to support the use of robotic-assisted bronchoscopy (RAB). Recent studies, which have been a mix of retrospective and small prospective trials, have reported a diagnostic yield between 80% and 93% with an excellent safety profile for shape-sensing RAB (ssRAB).^{9 12–16} In one study, the adjunctive use of cone beam CT (CBCT) imaging to overcome CT-to-body divergence increased the diagnostic yield from 29.9% to 86.6%.¹⁵

There is currently a lack of large prospective studies examining the role of ssRAB in the biopsy of small nodules and outcomes for nodules close to critical structures. In this trial, we report on the diagnostic and safety outcomes from the first European prospective multicentre trial which aims to assess the accuracy of ssRAB in accessing pulmonary nodules, with tool-in-lesion (TIL) confirmation by CBCT. Additionally, the relationship between outcomes and nodule characteristics (size and proximity to critical structures) will be explored. Tool-in-lesion was selected as the primary outcome to evaluate whether the ssRAB platform can accurately place a biopsy tool in the lesion. In addition, outcomes such as diagnostic yield can be affected by factors such as cancer prevalence, biopsy tool, histological processing and interpretation.

METHODS

Study cohort

200 subjects were enrolled consecutively into the trial from two thoracic centres between April 2023 and April 2024. The procedures were carried out by three operators with no prior experience in RAB (ie, 'RAB-naïve operators'). There was no run-in period and the study started from the first clinical cases. Adult patients with one or more suspicious pulmonary nodules requiring further investigation were referred by their multidisciplinary team or respiratory physician. The eligibility criteria were as follows: aged 18 years or older; suitable for elective nodule biopsy via bronchoscopy under general anaesthesia; solid or part solid nodules 6–30 mm in size and located at or distal to the fourth generation airway. Subjects were excluded if the target nodule was non-solid; if a bulla >1 cm was located near the target nodule or if the subject was unsuitable for a general anaesthetic procedure.

Study design

This is a single arm, multicentre prospective study evaluating the effectiveness and safety of ssRAB with adjunctive CBCT imaging for the biopsy of pulmonary nodules.

For each subject, the procedure pathway was created using a CT scan carried out within 30 days, on the Ion PlanPoint software (Intuitive Surgical). General anaesthesia was administered according to local protocol and all subjects were intubated. Conventional bronchoscopy was performed to assess for any endobronchial lesions and clear airway secretions. A swivel connector was attached to the end of the endotracheal tube and the robotic arm magnetically locked in place to this connector. Registration was performed to ensure that the virtual procedure pathway matches with the white light view. Once this was confirmed, the proceduralist navigated to the nodule. Augmented fluoroscopy (Phillips) and/or radial endobronchial

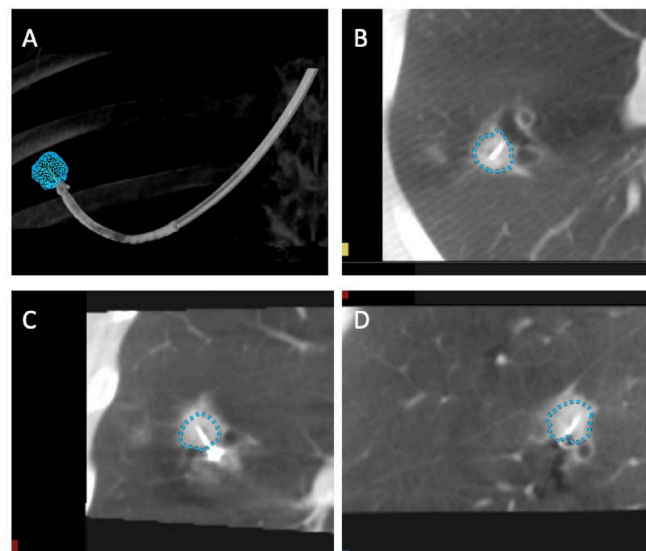


Figure 1 Cone beam CT imaging confirming tool-in-lesion Image. (A) Three-dimensional reconstruction; (B) axial view; (C) coronal view; (D) sagittal view.

ultrasound (Olympus) were used to confirm location prior to placing a biopsy needle. The position of the biopsy tool within the lesion was confirmed using fixed CBCT (Phillips Azurion, Phillips Allura) which was not integrated with the Ion endoluminal platform. A breath-hold (Positive end-expiratory pressure (PEEP) 5–20 cmH₂O) was performed prior to each CBCT spin with the aim of avoiding atelectasis, so the nodule could be seen on CBCT to allow evaluation of TIL.¹⁷ If there was evidence of atelectasis, recruitment manoeuvres were performed. Once TIL was confirmed, then sampling was performed with the biopsy needle and then with forceps. Biopsies were taken using 19G, 21G or 23G Flexision biopsy needles (Intuitive Surgical) and/or biopsy forceps (Captura Biopsy forceps, Cook Medical). In addition, according to operator discretion, brushes (Olympus) were used and/or a bronchoalveolar lavage was taken for cytology. A sufficient tissue sample was determined by the proceduralist. Rapid on-site evaluation (ROSE) was not available and cryobiopsy was not permitted in this study.

Subjects received postprocedure care in line with local hospital policy, with a chest radiograph performed at least 1 hour after the procedure and subjects were discharged on the same day, if clinically well.

All subjects were followed up at 7 and 30 days. Subjects with a non-malignant biopsy result had further follow-up at 6 months to allow for diagnostic accuracy determination.

Endpoints

The primary endpoint was TIL as confirmed by CBCT showing biopsy tool position within the target nodule. TIL was assessed by the proceduralist by review of the intraprocedural three-dimensional CBCT images in a minimum of two of the three cardinal planes (axial, coronal, sagittal), as shown in figure 1 and online supplemental materials).

Secondary endpoints included diagnostic yield, accuracy and sensitivity for malignancy using definitions from the American Thoracic Society/American College of Chest Physicians Delphi consensus paper. Biopsy outcomes were defined as malignant, specific benign, non-specific benign or non-diagnostic.^{18 19} For subject-level data, patients with multiple nodules were categorised

according to the result of the diagnostic nodule (ie, malignant or specific-benign). The diagnosis of each case was determined by a multidisciplinary panel.

Analysis of outcomes based on the diagnosis of the first nodule biopsied and per nodule was also performed.

Diagnostic outcomes were calculated using the following equations:

Diagnostic yield = (malignant+specific benign diagnoses)/total number of procedures performed

Diagnostic accuracy at 6 months = (true positive+true negative biopsies) / total number of procedures performed

Sensitivity for malignancy=true positive / (true positive+false negative biopsies)

Malignant biopsies were deemed as true positive. True negatives were determined as specific or non-specific benign biopsies confirmed as non-malignant after 6 months follow-up or further diagnostic procedure. False negatives were defined as non-malignant diagnoses (specific, non-specific or non-diagnostic) confirmed as malignant after 6 months follow-up or further diagnostic procedure.

Additional endpoints included procedure-related characteristics such as procedure time (defined as the time from catheter insertion to final removal) and imaging (imaging modality used, number of spins, radiation dose).

The predefined safety endpoint of the trial was the rate of procedure-related adverse events up to 30 days postprocedure. Adverse events were graded using CTCAE (Common Terminology Criteria for Adverse Events) definitions.²⁰

Statistical analysis

R V.4.4.1 or SAS V.9.4 software (SAS Institute) was used for all analyses. Discrete variables are described using frequency, a percentage with numerator, denominator and, if applicable, 95% exact CI. For continuous variables, depending on the distribution of the data, medians with IQRs or means with SD are reported. Normality was assessed by Shapiro-Wilk test. CIs for TIL are calculated using Clopper-Pearson exact method formula.

The data set was complete, with only minimal missing values which were considered negligible; therefore, no imputation was performed.

A generalised linear mixed model analysis was performed to assess the association between procedure or nodule characteristics and diagnostic yield as binary outcome (diagnostic vs non-diagnostic), both for univariate and multivariate analyses using a logistic model (fixed effect model was assumed). Variables in the multivariate analysis were selected based on previously published data including a systematic review by Ali *et al.*²¹ OR estimates are given with a 95% CI and p value (a p value<0.05 is considered statistically significant). As the analysis did not involve multiple comparisons, adjustments for multiplicity were not applicable. The outcome was diagnostic if the pathology confirmed either a malignant or specific benign diagnosis.

RESULTS

A total of 203 patients were screened and 200 subjects (100 participants in each of the two sites) were enrolled between April 2023 and April 2024 (table 1). Biopsies were performed in 198 subjects with 221 nodules sampled using ssRAB. Nearly all (196, 98%) subjects completed follow-up to 30 days and 54/56 (96%) subjects with non-malignant nodules completing follow-up to 6 months (figure 2). Of the six subjects who did not complete follow-up, five had chart reviews performed to assess for adverse

Table 1 Summary of subject demographics

Patient characteristics	
Participants	200
Age (years)	69 (62–74)
Gender	
Male	59 (29.5)
Female	141 (70.5)
BMI	25.3 (21.9–28.9)
Smoking status	
Former	99 (49.5)
Current	55 (27.5)
Never	46 (23)
History of cancer	
None	121 (60.5)
Thoracic	14 (7)
Extrathoracic	71 (35.5)
COPD	
Total	59 (29.5)
Chronic bronchitis	22 (11)
Emphysema	43 (21.5)
Frequent exacerbator	2 (1)
Interstitial lung disease/fibrosis	7 (3.5)
Family history of cancer	26 (13)
Brock Score	
Median (IQR)	24.7 (14.0–39.1)
<10%	34 (17)
≥10%	166 (83)
Method of nodule identification	
Incidental	112 (56)
Lung cancer screening programme	57 (28.5)
Follow-up for pulmonary symptoms/lung cancer suspicion	17 (8.5)
Other : Cancer surveillance programme	13 (6.5)
Not known	1 (0.5)
Data presented as counts with percentage in brackets unless stated otherwise. Median data presented with IQR in brackets.	
BMI, body mass index; COPD, chronic obstructive pulmonary disease.	

events and further investigations. One died 21 days after the procedure due to metastatic cancer.

The median nodule size was 13 mm with 12.1% of nodules being in contact with the pleura or fissure, and 40.2% located in the outer third of the lung. A bronchus sign was present in 26.8% of lesions. 23 patients had multiple nodules: 12 were unilateral, 11 were bilateral (including one patient with three nodules). Characteristics of the targeted nodules are in table 2.

Diagnostic outcomes

The rate of TIL (primary endpoint) was 99.0% (198/200). It was not achieved in two subjects due to technical issues. The tool was positioned in the inner third of the nodule for 92.4% and in the outer two-thirds for 7.6% of sampled nodules (table 3). Median number of CBCT spins to achieve TIL was 4 (inclusive of the study required spin to confirm TIL) with median combined radiation dose of 12.1 Gy.cm² (table 3).

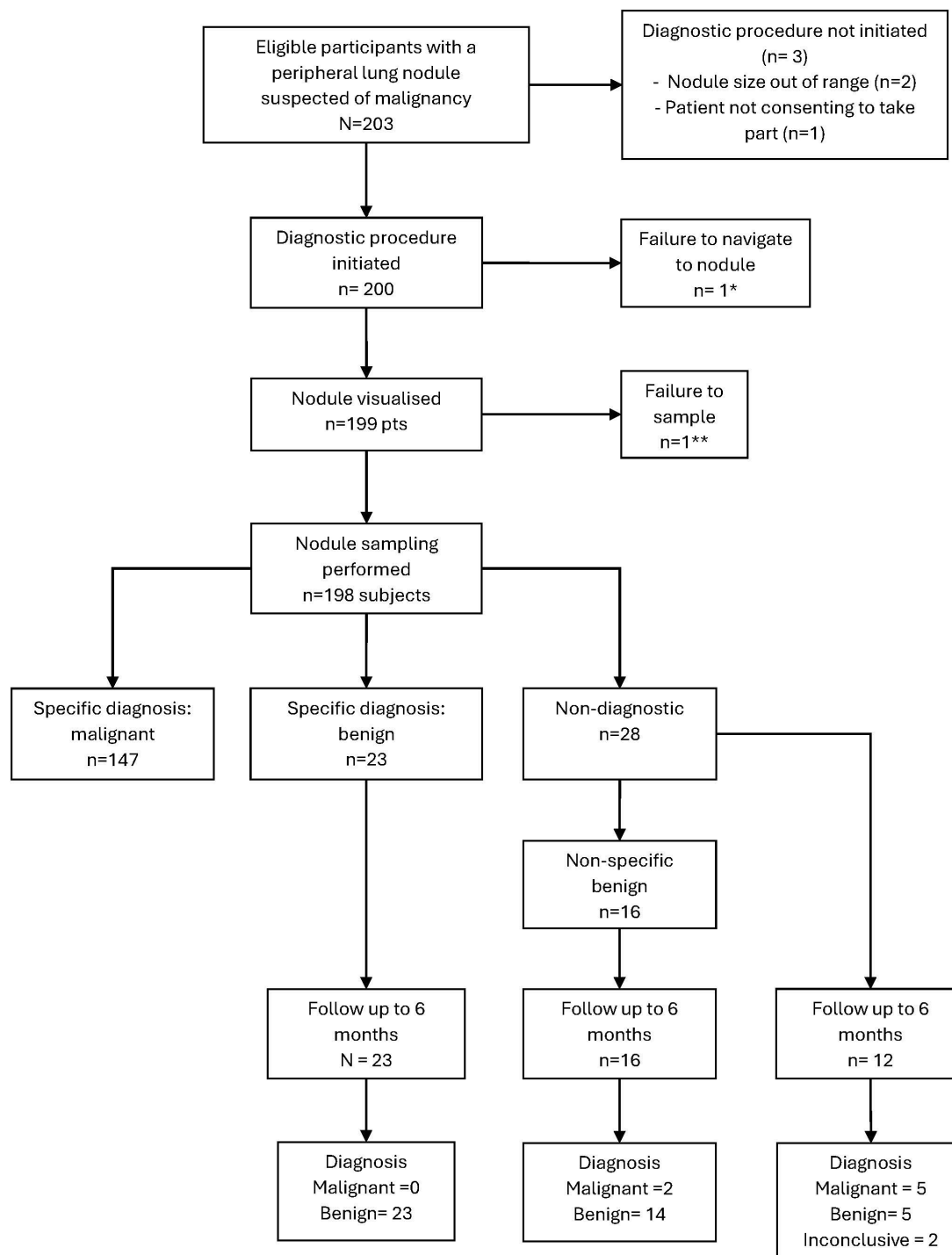


Figure 2 Flowchart describing subject selection and distribution of diagnoses immediately after biopsy and after 6 months of follow-up. Non-specific benign diagnoses include acute or chronic inflammation, fibrosis and suspected fibroelastosis. Non-diagnostic cases include non-specific benign and cases where navigation was attempted but no sample was obtained. Inconclusive: unable to confirm diagnosis due to lack of follow-up imaging or diagnostic procedure. *Navigation failed due to a technical issue; ** Tool in nodule could not be achieved due to a technical issue.

The histopathological outcomes were as follows: 147 malignant, 23 specific benign and 28 non-specific benign or non-diagnostic biopsies. Overall strict diagnostic yield was 85% (170/200). At 6 months, the diagnostic accuracy was 92.0% (184/200), sensitivity for malignancy was 95.5% (147/154) and the prevalence of cancer was 77.8% (154/198, [table 4](#)).

Diagnostic outcomes were also analysed per first nodule biopsied and per nodule. Per first nodule biopsied, the TIL

rate was 99% (198/200) with diagnostic yield of 83.5% (167/200). Per nodule, the TIL rate was 98.7% (221/224) and diagnostic yield was 81.3% (182/224, [online supplementary materials](#)).

A total of 87 nodules were confirmed as primary lung adenocarcinoma, 82 of which underwent molecular testing. In 70 (85.4%) cases, samples were sufficient for complete molecular analysis and in 7 (8.5%) sample adequacy only permitted partial molecular analysis.

Table 2 Summary of nodule characteristics

Nodule characteristics	
Total	224
Previous biopsy	
Yes	14 (6.2)
No	210 (93.8)
Biopsy performed	
Yes	221 (98.7)
No	3 (1.3)
Lesion size (mm)	
Median (IQR)	13 (10.9–16)
6–10 mm	41 (18.3)
10–20 mm	157 (70.1)
>20–30 mm	26 (11.6)
Lobar location	
Right upper lobe	72 (32.1)
Right middle lobe	23 (10.3)
Right lower lobe	44 (19.6)
Left upper lobe	59 (26.3)
Left lower lobe	26 (11.6)
Centrality	
Inner third	43 (19.2)
Middle third	91 (40.6)
Outer third	90 (40.2)
Nodule type	
Solid	182 (81.3)
Part solid	42 (18.8)
Bronchus sign	
Yes	60 (26.8)
No	164 (73.2)
Subpleural nodule	27 (12.1)
Median distance from pleura (IQR, mm)	19.9 (10–30)
Median distance from fissure (IQR, mm)*	25.7 (13–46)
Median distance to critical structure (IQR, mm)	35.0 (16.9–49.6)
Data presented as number with percentage in brackets unless stated otherwise. Lesion size is based on longest diameter as measured on CT. Distance to pleura/fissure/critical structure as measured on PlanPoint software. Bronchus sign is defined as an airway leading directly to a nodule on CT imaging. Critical structure such as heart, aorta, main pulmonary vessels or other central blood vessels. Subpleural nodule: nodule in contact with the pleura or fissure.	
*Unable to measure for one nodule.	

Sensitivity and post hoc analyses

If the two inconclusive cases were malignant on follow-up, then the sensitivity for malignancy in this trial would be 94.2% (147/156). Conversely, if all cases were benign, then sensitivity would remain at 95.5%. The outcomes of the two cases would have no impact on diagnostic yield or accuracy.

Outcomes according to nodule location or size

Post hoc analysis of diagnostic outcomes for ssRAB with CBCT in smaller nodules and nodules located within 5 mm of the pleura (peripleural), fissure (perifissural) or a critical structure (eg, paramediastinal—heart, aorta or main pulmonary artery) was

Table 3 Summary of procedure outcomes and rate of tool-in-lesion

Procedure characteristics	
Total number of procedures	200
Procedure time* (minutes; median, IQR)	
Overall	44 (33–56)
First 10 subjects at each site	59.0 (52.5–91.5)
Last 10 subjects at each site	35.0 (30.5–53.5)
Total in-room time† (minutes; median, IQR)	
Overall	74 (60–90)
First 10 subjects at each site	97 (83–122)
Last 10 subjects at each site	63 (52–78)
Number of nodules per attempted procedure	
1	177 (88.5)
2	22 (11)
3	1 (0.5)
Intraoperative imaging per nodule	
Radial EBUS	112 (50.4)
Concentric	78 (69.6)
Eccentric	33 (29.5)
Not visualised	1 (0.9)
Cone beam CT	224 (100)
Fluoroscopy	224 (100)
Median radiation dose (IQR — DAP, Gy.cm ²)	
Fluoroscopy	2.5 (1.2–4.8)
Cone beam CT	8.3 (4.9–16.0)
Combined	12.1 (7.6–19.9)
Median number of spins (IQR)	4 (3–7)
Median fluoroscopy time (minutes, IQR)	6.38 (3.78–9.38)
Tool-in-lesion outcomes	
TIL achieved	
Yes (%; n/N)‡	198 (99.0)
95% CI	96.05 to 99.83
Location of tool in lesion	
Inner 1/3 of nodule	183 (92.4)
Outer 2/3 of nodule	15 (7.6)
Distance from catheter tip to nodule edge (3D imaging n=222)‡	
Median (IQR)	0.8 mm (0–3.4 mm)
Data presented as number with percentage in brackets unless stated otherwise. *Procedure time = difference between robotic catheter insertion and final removal time. †In-room time = difference between room in and room out time; data missing for 1 patient (part of the last 10 subjects subgroup). ‡In one patient (two nodules), navigation failed due to a technical issue; in another patient TIL could not be achieved due to a technical issue. For the first patient, the distance from catheter tip to the two nodules is not available as navigation was aborted prior to reaching the nodule. 3D, three-dimensional; DAP, Dose area product; EBUS, endobronchial ultrasound; TIL, tool-in-lesion.	

performed (figure 3). The diagnostic accuracy for peripleural and perifissural was 93.3% (56/60), and for paramediastinal nodules was 100% (11/11) (figure 4). For nodules under 15 mm and 20 mm, the diagnostic accuracies were 86.1% (124/144) and 88.2% (172/195) (figure 4). There was no significant difference in the median number of CBCT spins required to achieve TIL.

Table 4 Summary of pathology outcomes per procedure at point of biopsy and after 6 months of follow-up

Pathology outcomes		
Malignant	Adenocarcinoma	84
	Typical carcinoid	19
	Metastatic cancer	17
	Squamous cell carcinoma	14
	Non-small cell cancer - not otherwise specified	6
	Lymphoma	2
	Dual cancers	2
	Small cell carcinoma	1
	Preinvasive lesion	1
	Adenosquamous	1
	Specific benign	Granulomatous disease
Hamartoma		3
Schwannoma		1
Chondroma		1
Bacterial		4
Fungal		1
Organising pneumonia		2
Lymph node		1
Amyloidosis		1
IgG4 disease		1
Non-diagnostic		Atypical cells
	Normal respiratory tissue	8
	Acute/chronic inflammation, Fibrosis¶	15
	Fibroelastosis (suspected)¶	1
	Specimen/tissue inadequacy	1
6 month follow-up outcomes	TP*	147
	FP	0
	TN†	37
	FN‡	7
	Inconclusive§	4
Diagnostic yield	$\frac{Malignant + SpB}{All\ procedures} = \frac{147 + 23}{200} = 85.0\%$	
Diagnostic accuracy	$\frac{TP + TN(SpB + NSpB)}{All\ procedures} = \frac{147 + 37}{200} = 92.0\%$	
Sensitivity for malignancy	$\frac{TP}{TP + FN} = \frac{147}{147 + 7} = 95.5\%$	
Cancer prevalence	$\frac{TP + FN}{All\ Biopsied\ Patients} = \frac{147 + 7}{198} = 77.8\%$	
*All malignant biopsies were deemed TP at 6 months.		
†TN: specific or non-specific benign biopsies confirmed as non-malignant after 6 months follow-up or further diagnostic procedure.		
‡FN: non-malignant diagnoses (specific, non-specific or non-diagnostic) confirmed as malignant after 6 months follow-up or further diagnostic procedure.		
§Inconclusive: unable to confirm diagnosis due to lack of follow-up imaging or diagnostic procedure; includes two procedures where navigation was attempted but no sample was obtained.		
¶Non specific benign diagnoses		
FN, false negative; FP, false positive; NSpB, non-specific benign; SpB, specific benign; TN, true negative; TP, true positive.		

Univariate and multivariate analyses were performed (figure 5). Nodule type and centrality were not predictors for diagnostic yield. Univariate analysis showed statistically significant higher diagnostic yield for nodules with a positive bronchus sign (OR 2.87, 95%CI 1.157 to 8.733, $p=0.036$; Wald z-test); however, it was not significant on multivariate analysis (OR



Figure 3 CT images of paramediastinal, perifissural and peripleural nodules Top: axial view of 8.6×10×12.4 mm right middle lobe nodule, confirmed carcinoid. Middle: coronal view of 9.4×10.1×13 mm right middle lobe nodule, confirmed adenocarcinoma. Bottom: axial view of 7.6×8.9×8.5 mm left upper lobe nodule, confirmed adenocarcinoma.

1.807, 95%CI 0.673 to 5.755, $p=0.270$; Wald z-test). Nodule size was a significant positive predictor in both models.

Safety outcomes

Two subjects had episodes of bleeding requiring balloon tamponade. One case of pneumothorax was reported in a patient whose nodule was located on the diaphragm, for an overall rate of 0.5% (1/200), and was treated with a chest drain. There were four procedure-related serious adverse events (table 5) and the overall procedure-related serious adverse event rate was 2% (4/200) which was inclusive of the pneumothorax requiring chest drain. There were no device-related or procedure-related deaths. One subject died due to metastatic cancer (see online supplemental materials).

DISCUSSION

To our knowledge, this is the largest European multicentre prospective trial to date assessing the rate of TIL and diagnostic

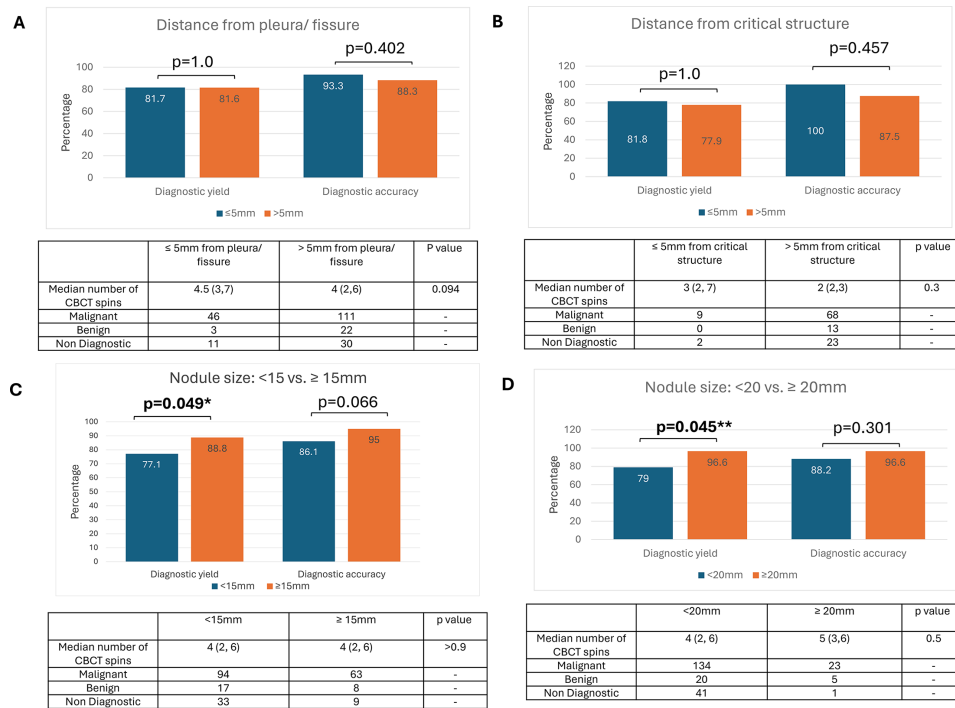


Figure 4 (A–D) Bar graph and tables showing diagnostic outcomes and median number of CBCT spins to achieve TIL according to nodule size and locations. *Effect size=0.314; **Effect size=0.579 per Cohen's *h*; *p* values: χ^2 . (A) Outcomes according to distance from pleura or fissure; (B) Outcomes according to distance from critical structure; (C) Nodules under or over 15 mm; (D) Nodule size under or over 20 mm. CBCT, cone beam CT; TIL, tool-in-lesion.

yield of ssRAB used in combination with CBCT. We report a TIL rate of 99.0% (198/200) with a strict diagnostic rate of 85% (170/200), diagnostic accuracy of 92.0% (184/200) at 6 months and sensitivity for malignancy of 95.5% (147/154). Molecular analysis was fully or partially adequate in 93.9% of adenocarcinoma cases. The targeted nodules were small, when compared with previous studies, with a median size of 13 mm. A significant proportion (80.8%) were located within the peripheral lung.

Our TIL rate was 99.0% which supports the hypothesis that ssRAB with CBCT enables precise placement of tools within the target lesion. TIL was chosen as the primary end point as other measures such as diagnostic yield are also determined by confounding factors including cancer prevalence and choice of biopsy technique such as specific tools and ROSE.²² TIL is also a relevant outcome to assess the ability of ssRAB to deliver therapeutics which is a recent development in this field.

The diagnostic outcomes from this trial compare favourably to previous studies. Ali *et al* published a meta-analysis of 13 ssRAB trials, with and without cross-sectional adjunctive imaging, and reported overall pooled diagnostic yields of 87.2% and 85.2%, respectively.²¹ Notably, this meta-analysis included follow-up data when calculating diagnostic yields, and the mean nodule size ranged from 14 to 22.6 mm.

Our findings confirm outcomes from a retrospective comparative study by Abdelghani *et al* which evaluated EMN with CBCT and ssRAB with CBCT. They reported a diagnostic yield of 66% for EMN-CBCT and 89.2% for ssRAB-CBCT. The median nodule size in the ssRAB-CBCT group was 12 mm and bronchus sign was present in 33% of nodules.¹⁴

We also analysed outcomes per first nodule biopsied and per nodule with diagnostic yields of 83.5% and 81.3%, respectively. These diagnostic outcomes are similar to results from the Robotic versus Electromagnetic Bronchoscopy for Pulmonary

Lesion Assessment (RELIANT) trial which reported a diagnostic yield of 76.4% for RAB group and 75% in EMN group.²³

A significant challenge for navigation bronchoscopy is CT-to-body divergence as the pulmonary environment changes when a patient is under anaesthesia with mechanical ventilation, relative to their awake state.²⁴ Navigation procedures are planned using a preprocedural CT scan performed at full inspiration; however, the position of the target nodule may be changed through any combination of displacement by atelectasis, diaphragmatic movement and vascular pulsation. Our adjunctive use of CBCT and augmented fluoroscopy helps to overcome these issues, therefore improving TIL rate and diagnostic outcomes.^{15 25}

We showed that nodules located within 5 mm from critical structures or the pleura/fissure can be safely biopsied with a diagnostic yield of 82% and accuracy between 93.3% and 100%. This was achievable as the ssRAB platform provided peripheral stability and adjunctive imaging (CBCT, fluoroscopy) provided real-time confirmation of both catheter and biopsy tool position. To date, there has only been a single retrospective trial investigating nodules abutting pleura or fissures which reported a diagnostic yield and accuracy of 80.2%. Of note, the nodules were larger (median size 19 mm); more nodules had a bronchus sign (50%) and cryobiopsies were used in 47.8% of cases.⁹

Currently, CT-TTNB remains the gold standard for investigating pulmonary lesions with a pooled diagnostic yield of 93% for nodules under 3 cm.²⁶ However, in a prospective series, the strict diagnostic yield was 78% in 319 procedures with a median nodule size of 23.2 mm.^{26 27} Additionally, a multicentre retrospective trial comparing ssRAB with CT-TTNB reported a diagnostic accuracy of 87.6% and 88.4%, respectively.²⁸ The NaVigation Endoscopy to Reach Indeterminate lung nodules versus Trans-thoracic needle Aspiration (VERITAS) study reported diagnostic accuracy of 79% in the EMN group and 73.6% in the CT-TTNB

Variable	Univariate analysis		Multivariate analysis	
	Estimate OR (95% CI)	P value	Estimate OR (95% CI)	P value
Location				
Upper	0.973 (0.447, 2.018)	0.943	1.257 (0.550, 2.778)	0.575
Lower	...	...	...	...
Nodule Type				
Solid	0.93 (0.356, 2.194)	0.887	0.827 (0.299, 2.054)	0.697
Semisolid	...	...	...	...
Nodule Size ^[1]	1.169 (1.066, 1.302)	0.002*	1.147 (1.042, 1.281)	0.008*
Lung Centrality				
Inner 1/3	1.650 (0.635, 4.849)	0.326	1.774 (0.619, 5.898)	0.310
Middle 1/3	1.471 (0.690, 3.202)	0.320	1.431 (0.627, 3.327)	0.396
Outer 1/3	...	...	...	...
Bronchus sign				
Present ^[2]	2.87 (1.157, 8.733)	0.036*	1.807 (0.673, 5.755)	0.270
Absent	...	...	...	...
Pleura wall or Fissure				
>5mm	0.964 (0.419, 2.073)	0.928	0.992 (0.407, 2.284)	0.986
≤5mm	...	...	...	...

Figure 5 Univariate and multivariate predictors for diagnostic yield. OR estimates are given with 95% CI and p value (*a p value<0.05 was considered statistically significant; Wald z-test) [1] Longest nodule diameter; [2] Present=Airway directly leading to nodule; Absent=Absent or airway adjacent to the nodule.

group.²⁹ Consequently, these studies, together with our findings here, provide a growing body of evidence that the diagnostic yield of navigation bronchoscopy, and in particular ssRAB, is at least as good, if not superior to CT-TTNB.

We reported a serious adverse event rate of 2%, and pneumothorax rate of 0.5% (following biopsy of a nodule adjacent to the diaphragm). In comparison, a meta-analysis of CT-TTNB had a pooled pneumothorax rate of 25.3%.³⁰

As three-dimensional imaging techniques become more widely used, particularly in lung cancer screening and other non-thoracic conditions, a greater number of pulmonary nodules of varying sizes and locations will be detected. Many of these

nodules will be challenging to biopsy, particularly those adjacent to the heart or diaphragm.⁵ Our study data suggests that ssRAB in combination with CBCT can be delivered safely, and has as good a diagnostic yield as nodules not near to critical structures.

The use of CBCT scans and fluoroscopy raises the concern of radiation risk. In this trial, the median total radiation dose was 12 Gy.cm² which is equivalent to an effective dose of 1.92 milliSieverts (mSv) using a tissue weighting factor of 0.16 mSv/Gy.cm².^{31,32} Notably, the radiation dose for a standard CT scan of the thorax is 8 mSv.³³ It is challenging to draw direct comparisons to the radiation dose of CT-TTNB as they are typically reported as a mean value with reported values between 0.5 and 7.3 mSv.^{34–36} In addition, the primary end point of our trial was TIL which mandates at least one CBCT spin to confirm this and therefore the reported radiation will likely be higher than that in non-trial cases.

Among the 28 patients with non-diagnostic and non-specific benign index diagnoses, 19 were confirmed to be benign at 6 month follow-up and 2 (non-diagnostic) subjects are awaiting follow-up imaging with no evidence of malignancy on clinical assessment. There were seven false negative cases; however, in two cases the biopsies found atypical cells which were suspicious enough to warrant surgical resection or rebiopsy with RAB. The outcomes of the two inconclusive cases would have no impact on diagnostic yield or accuracy.

Table 5 Summary of key procedure-related adverse events

Key procedure-related adverse events	N, (%)	CTCAE grade
Bleeding Nashville grade ≥3	2 (1)	2
Pneumothorax*	1 (0.5)	3
Procedure-related serious adverse events		
Bronchial infection	1 (0.5)	3
Sinus bradycardia	1 (0.5)	3
Suspected embolism / cerebrovascular accident	1 (0.5)	4

*Pneumothorax required chest drain and overnight admission therefore was a serious adverse event but not listed twice to avoid duplication.
CTCAE, Common Terminology Criteria for Adverse Events.

A limitation of our study is the involvement of RAB-naïve operators (but who have previous navigational bronchoscopy experience), as well as the lack of use of integrated CBCT, cryobiopsy and ROSE, all of which could underestimate the diagnostic performance of ssRAB. This may, in part, explain the discrepancy between the achieved TIL rate and the reported diagnostic outcomes. Furthermore, a cancer prevalence of 77.8% will have impacted the diagnostic yield.

The use of TIL as a primary outcome could be seen as a limitation of this study due to lack of technique standardisation, validation and potential bias with the proceduralist determining success. We would, however, argue that it is an important outcome to report on as it measures two critical components of a successful biopsy: navigation and sampling location. Furthermore, the proceduralist determining successful TIL reflects real-world practice as operators themselves would need to decide if they have reached the target nodule. Notably, studies have shown that achieving TIL improves diagnostic yield with an increased odds of 53 times for CBCT guided transbronchial biopsies.^{37,38} Our method of TIL confirmation is similar to previous studies^{39,40} and our outcomes are comparable to those reported by Bashour *et al.*¹⁵

Conversely, the strength of this study is in the prospective, multicentre design in novice RAB sites with no experience in this procedure, so the findings can be generalised to navigational bronchoscopy centres developing a ssRAB service.

In summary, TIL can be reliably achieved in small and peripleural, perifissural and paramedistinal pulmonary nodules with ssRAB and CBCT. Despite the difficulties in comparing diagnostic outcomes between biopsy techniques, the results of this study suggest that the diagnostic performance of ssRAB with CBCT is at least comparable to the current standard of care (CT-TTNB) with the known advantages of a bronchoscopic approach. ssRAB can be performed safely in patients who are at high risk of pneumothorax and for patients with multiple targets.

Acknowledgements The authors would like to thank all trial subjects for their participation. They would also like to thank the bronchoscopy, anaesthetic, radiology and theatre staff for their assistance in performing the procedures. Additionally, the authors would like to thank the Intuitive team for their role in designing the trial, monitoring study data and analysis of results.

Contributors LTC: data collection, screening of articles, writing—original draft; writing—review and editing. KKWL: study conception and design, project supervision, writing—original draft, writing—review and editing. CMO, KT, AT, IB, AK, EA, JG, AK, ML, FT: data collection, writing—review and editing. PLS: study conception and design, project supervision, writing—original draft, writing—review and editing. PLS is the guarantor.

Funding Intuitive Surgical funded the trial.

Competing interests Intuitive Surgical is the sponsor for the study. PLS completed paid lectures for Intuitive Surgical. CMO completed paid lectures and consulting including travel fees for Intuitive Surgical. KKWL completed paid lectures and consulting for Intuitive Surgical. KKWL is a member of their advisory board.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by East of England—Cambridge East REC ref 22/EE/0271. Participants gave informed consent to participate in the study before taking part. The study was conducted in line with local regulatory requirements and the Declaration of Helsinki.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information.

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REFERENCES

- Field JK, Duffy SW, Baldwin DR, *et al.* UK Lung Cancer RCT Pilot Screening Trial: baseline findings from the screening arm provide evidence for the potential implementation of lung cancer screening. *Thorax* 2016;71:161–70.
- Wilson DO, Weissfeld JL, Fuhrman CR, *et al.* The Pittsburgh Lung Screening Study (PLUSS): outcomes within 3 years of a first computed tomography scan. *Am J Respir Crit Care Med* 2008;178:956–61.
- Das A, Bonney A, Manser R. Prevalence of pulmonary nodules detected incidentally on noncancer-related imaging: a review. *Intern Med J* 2024;54:1440–9.
- Najafi G, Lakshminarayan R, Haslam P, *et al.* Interventional radiology procedures, facilities, and workforce across England and Wales: a snapshot retrospective evaluation from 2017 to 2021. *Clin Radiol* 2024;79:e1481–9.
- Kemp SV. Navigation Bronchoscopy. *Respiration* 2020;99:277–86.
- Yeow K-M, Su I-H, Pan K-T, *et al.* Risk factors of pneumothorax and bleeding: multivariate analysis of 660 CT-guided coaxial cutting needle lung biopsies. *Chest* 2004;126:748–54.
- de Koning HJ, van der Aalst CM, de Jong PA, *et al.* Reduced Lung-Cancer Mortality with Volume CT Screening in a Randomized Trial. *N Engl J Med* 2020;382:503–13.
- Horeweg N, van der Aalst CM, Thunnissen E, *et al.* Characteristics of lung cancers detected by computer tomography screening in the randomized NELSON trial. *Am J Respir Crit Care Med* 2013;187:848–54.
- Fernandez-Bussy S, Yu Lee-Mateus A, Barrios-Ruiz A, *et al.* Diagnostic performance of shape-sensing robotic-assisted bronchoscopy for pleural-based and fissure-based pulmonary lesions. *Thorax* 2025;80:150–8.
- Hsu LS, Chen CW, Chang CH, *et al.* Influences of cardiac motion on computed tomography-guided biopsy of lung nodules located near the heart. *Medicine (Baltimore)* 2017;96:e8558.
- Folch EE, Pritchett MA, Nead MA, *et al.* Electromagnetic Navigation Bronchoscopy for Peripheral Pulmonary Lesions: One-Year Results of the Prospective, Multicenter NAVIGATE Study. *J Thorac Oncol* 2019;14:445–58.
- Abia-Trujillo D, Funes-Ferrada R, Yu Lee-Mateus A, *et al.* Cryobiopsy versus fine-needle aspiration for shape-sensing robotic-assisted sampling of small lung nodules. *Lung Cancer (Auckl)* 2024;196:107967.
- Abia-Trujillo D, Folch EE, Yu Lee-Mateus A, *et al.* Mobile cone-beam computed tomography complementing shape-sensing robotic-assisted bronchoscopy in the small pulmonary nodule sampling: A multicentre experience. *Respirology* 2024;29:324–32.
- Abdelghani R, Espinoza D, Uribe JP, *et al.* Cone-beam computed tomography-guided shape-sensing robotic bronchoscopy vs. electromagnetic navigation bronchoscopy for pulmonary nodules. *J Thorac Dis* 2024;16:5529–38.
- Bashour SI, Khan A, Song J, *et al.* Improving Shape-Sensing Robotic-Assisted Bronchoscopy Outcomes with Mobile Cone-Beam Computed Tomography Guidance. *Diagnostics (Basel)* 2024;14:1955.
- Kalchiem-Dekel O, Connolly JG, Lin I-H, *et al.* Shape-Sensing Robotic-Assisted Bronchoscopy in the Diagnosis of Pulmonary Parenchymal Lesions. *Chest* 2022;161:572–82.
- Pritchett MA, Lau K, Skibo S, *et al.* Anesthesia considerations to reduce motion and atelectasis during advanced guided bronchoscopy. *BMC Pulm Med* 2021;21:240.
- Gonzalez AV, Silvestri GA, Korevaar DA, *et al.* Assessment of Advanced Diagnostic Bronchoscopy Outcomes for Peripheral Lung Lesions: A Delphi Consensus Definition of Diagnostic Yield and Recommendations for Patient-centered Study Designs. An Official American Thoracic Society/American College of Chest Physicians Research Statement. *Am J Respir Crit Care Med* 2024;209:634–46.
- Vachani A, Maldonado F, Laxmanan B, *et al.* The Impact of Alternative Approaches to Diagnostic Yield Calculation in Studies of Bronchoscopy. *Chest* 2022;161:1426–8.
- Services USHhA. Common terminology criteria for adverse events (CTCAE) version 5.0. 2017.
- Ali MS, Ghori UK, Wayne MT, *et al.* Diagnostic Performance and Safety Profile of Robotic-assisted Bronchoscopy: A Systematic Review and Meta-Analysis. *Ann Am Thorac Soc* 2023;20:1801–12.

- 22 Vachani A, Maldonado F, Laxmanan B, *et al.* The Effect of Definitions and Cancer Prevalence on Diagnostic Yield Estimates of Bronchoscopy: A Simulation-based Analysis. *Ann Am Thorac Soc* 2023;20:1491–8.
- 23 Paez R, Lentz RJ, Duke JD, *et al.* Robotic versus Electromagnetic Bronchoscopy for Peripheral Pulmonary Lesions: A Randomized Trial (RELIANT). *Am J Respir Crit Care Med* 2025;211:1644–51.
- 24 Chan LT, Orton CM, Shah PL. A New Dawn for Bronchoscopy for Peripheral Lung Lesions?. *Chest* 2024;166:931–3.
- 25 Abdelghani R, Omballi M, Abia-Trujillo D, *et al.* Imaging modalities during navigational bronchoscopy. *Expert Rev Respir Med* 2024;18:175–88.
- 26 Han Y, Kim HJ, Kong KA, *et al.* Diagnosis of small pulmonary lesions by transbronchial lung biopsy with radial endobronchial ultrasound and virtual bronchoscopic navigation versus CT-guided transthoracic needle biopsy: A systematic review and meta-analysis. *PLoS One* 2018;13:e0191590.
- 27 Tipaldi MA, Ronconi E, Krokidis ME, *et al.* Diagnostic yield of CT-guided lung biopsies: how can we limit negative sampling? *Br J Radiol* 2022;95:20210434.
- 28 Yu Lee-Mateus A, Reisenauer J, Garcia-Saucedo JC, *et al.* Robotic-assisted bronchoscopy versus CT-guided transthoracic biopsy for diagnosis of pulmonary nodules. *Respirology* 2023;28:66–73.
- 29 Lentz RJ, Frederick-Dyer K, Planz VB, *et al.* Navigational Bronchoscopy or Transthoracic Needle Biopsy for Lung Nodules. *N Engl J Med* 2025;392:2100–12.
- 30 Heerink WJ, de Bock GH, de Jonge GJ, *et al.* Complication rates of CT-guided transthoracic lung biopsy: meta-analysis. *Eur Radiol* 2017;27:138–48.
- 31 Pritchett MA, Schampaert S, de Groot JAH, *et al.* Cone-Beam CT With Augmented Fluoroscopy Combined With Electromagnetic Navigation Bronchoscopy for Biopsy of Pulmonary Nodules. *J Bronchology Interv Pulmonol* 2018;25:274–82.
- 32 Styrvoky K, Schwalk A, Pham D, *et al.* Radiation dose of cone beam CT combined with shape sensing robotic assisted bronchoscopy for the evaluation of pulmonary lesions: an observational single center study. *J Thorac Dis* 2023;15:4836–48.
- 33 England PH. Patient dose information: guidance. 2008. Available: <https://www.gov.uk/government/publications/medical-radiation-patient-doses/patient-dose-information-guidance>
- 34 Guberina N, Forsting M, Ringelstein A, *et al.* Radiation exposure during CT-guided biopsies: recent CT machines provide markedly lower doses. *Eur Radiol* 2018;28:3929–35.
- 35 Kim GR, Hur J, Lee SM, *et al.* CT fluoroscopy-guided lung biopsy versus conventional CT-guided lung biopsy: a prospective controlled study to assess radiation doses and diagnostic performance. *Eur Radiol* 2011;21:232–9.
- 36 Fu YF, Li GC, Xu QS, *et al.* Computed tomography-guided lung biopsy: a randomized controlled trial of low-dose versus standard-dose protocol. *Eur Radiol* 2020;30:1584–92.
- 37 Park SC, Kim CJ, Han CH, *et al.* Factors associated with the diagnostic yield of computed tomography-guided transbronchial lung biopsy. *Thorac Cancer* 2017;8:153–8.
- 38 Casal RF, Sarkiss M, Jones AK, *et al.* Cone beam computed tomography-guided thin/ultrathin bronchoscopy for diagnosis of peripheral lung nodules: a prospective pilot study. *J Thorac Dis* 2018;10:6950–9.
- 39 Reisenauer J, Duke JD, Kern R, *et al.* Combining Shape-Sensing Robotic Bronchoscopy With Mobile Three-Dimensional Imaging to Verify Tool-in-Lesion and Overcome Divergence: A Pilot Study. *Mayo Clin Proc Innov Qual Outcomes* 2022;6:177–85.
- 40 Bhadra K, Rickman OB, Mahajan AK, *et al.* "Tool-in-lesion" Accuracy of Galaxy System-A Robotic Electromagnetic Navigation Bronchoscopy With Integrated Tool-in-lesion-Tomosynthesis Technology: The MATCH Study. *J Bronchology Interv Pulmonol* 2024;31:23–9.